

Compressor Rotor Joints and Fits Interference Study

Preliminary Design Review

by

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Four Panel Chart

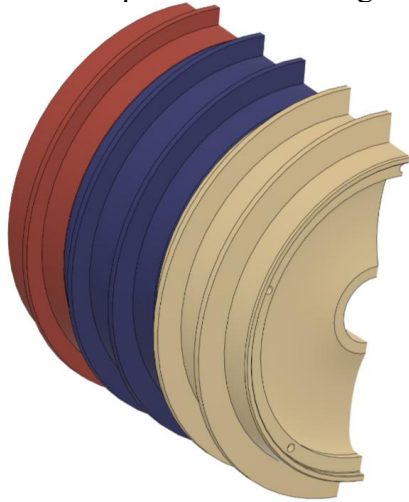
Project Overview Solar Turbines needs a comprehensive study tool on the effects axial and radial interference fits have on turbine compressor rotor stiffness. The weldments of the compressor have press-fit dowel pins and pilot joints to connect one weldment section to another. These fits need to be assessed through a finite element modal analysis as well as practically tested through modal impacts on manufactured prototypes to investigate how the fits affect the stiffness of the turbine's connected weldments. The experimental data will be used to validate the finite element analysis (FEA) model tool to eventually be presented to Solar Turbines.	Concept description Our Concept CAD is comprised of 3 individual components: aft hub, aft weldment, and forward weldment. These sections will be connected and analyzed on multiple different mating techniques. 
Concept justification The software we are using for our turbine analysis is Ansys. We are using Ansys because it is highly capable for finite element studies and is widely used in industry. In addition, our sponsor, Solar Turbines, emphasizes that the analysis to be with Ansys. Regarding our physical scaled down prototype, we are going to select pin and interference configurations that our finite element analysis determines to be most optimal for increasing assembly stiffness. The reason for this selection is to first validate that our finite element analysis can accurately predict physical characteristics like natural frequency and effective stiffness. We will then test the different configurations on the physical prototype to validate that the optimization technic utilized in our software matches real world results.	What's Next The next step for our project includes beginning development an ANSYS model that factors in thermal effects and contact stress to accurately solve for the natural frequencies of the design. The goal is to have a sufficient model by the end of winter break. We will also determine a schedule for when the prototype models should be accomplished. Other steps include repeated internal designs reviews, feasibility prototypes, and creating engineering drawings to prepare for manufacturing.

Figure 1. Four-Panel Chart for Preliminary Design Review

Overview

The system we will create to analyze the turbine compressor will consist of an accurate Ansys

simulation model used to determine the optimal press-fit types and configurations to best increase rotor stiffness. Our plan is to create an Ansys model that Solar Turbines can use to study these different configurations and press fit interferences to use in further design decisions. The general workflow of this Ansys model can be visualized in Figure 2z below. After studying the load paths and rotor dynamics of the Titan 130 gas turbine, we will study the methods of stress analysis with Ansys and create a physical model of the forward, aft, and aft hub weldments to find the best proposed mating technique. Our final product will be the complete Ansys model with a report on our findings and test data from interference fits, making an overall tool Solar Turbines can use in the future.

We plan to use the tool we make to find the best three mating configurations from Ansys to present to Solar Turbines. To validate our Ansys model, we plan to test the tool on a scaled-down physical model that will be machined out of steel for validation. So far, we have developed Pugh and decision matrices on different press fit configurations we will run in the Ansys simulation. From the Ansys simulation, we will find the optimal top three to test physically. We will then conceive a specific plan to test those three configurations on our physical model and refine any inaccuracies in the Ansys model. Validation of our ANSYS model will come through comparing the deflection and natural frequencies measured from the test set up, hand calculations, and having a percent error no greater than 10%.

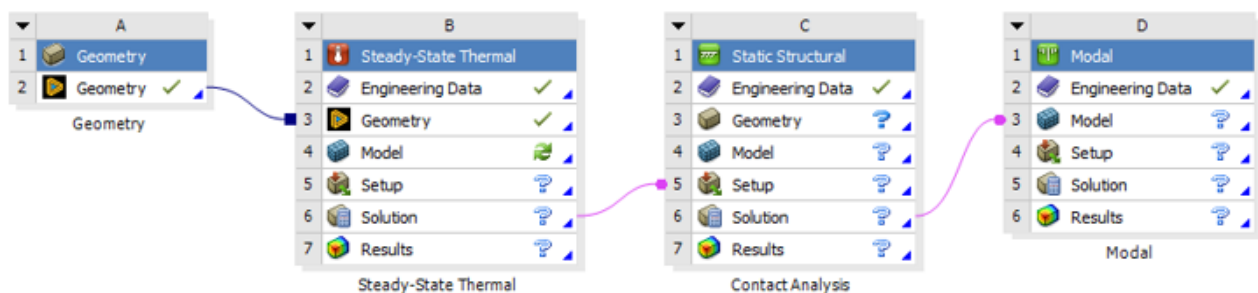


Figure 2. Ansys workflow from modeling physical geometry to finding results.

This report will cover our concept ideas for the Ansys simulation tool in greater detail, as well as the next steps in studying the interference fits and manufacturing the physical model. We will use this report to explain our intentions and ideas for this project, as well as gain feedback from our coaches and our sponsor about our research methods. This project overall is research focused and thus needs validation on whether our plans to create an Ansys study tool will be sufficient to meet Solar Turbine's needs. In the appendix of this report, we include various decision matrices, a preliminary analysis, and our project Gantt chart to provide resources on our overall design process so far. The decision matrices of appendices A and B were used to organize the many different mating methods that could be used for the rotors into reasonable options to be studied with the Ansys model. The preliminary analysis shown in appendix C was an initial mathematical analysis on the forces presented in the pins during the compressor's operation. The Gantt chart in appendix D shows our project timeline in the following weeks up to the CDR.

Concept Description and Justification

Considering the nature of our project, it is difficult to predict the exact interference fits and

configurations that our Ansys tool will consider the most optimal for increasing rotor stiffness. Thus, we will begin by analyzing the most common and widely used mating methods used in turbine engines. There is no exact certainty that our software will align with these current mating techniques being the most optimal, but to talk about different concepts for our project, we are forced to make certain assumptions. Those assumptions include that the most likely configurations that we are going to be testing and manufacturing are going to be a 5 and 7 dowel pin configuration with different pilot and dowel pin interference fits. We believe that these assumptions are relatively reasonable because they are tested configurations that are used in other turbines, giving these configurations merit and a decent probability that they will have similar performance characteristics to Solar Turbines' current configuration as shown in Figure 2. This helps us establish a safe and realistic starting point of what configurations work and which direction we can move our design to increase stiffness and performance characteristics.

In order to have a deeper understanding of the complexity and decision making regarding our project, we first must emphasize some driving engineering requirements that must be met by our design. First, Solar Turbines has a 30,000-hour operating time requirement for which their turbines must operate without any maintenance or inspection. To meet this requirement, it must be decomposed into several child requirements that must be met prior to make this requirement feasible. Some of those child requirements being that the mating method must not shear or slip during operation. In addition, the mating method must be sufficient at the operating temperature and pressure. Consequently, the mating surface must also not reach fatigue failure before the 30,000-hour requirement given by Solar Turbines. Another important requirement that must be met by our design is that it must be manufacturable and installable, preventing any highly advanced or extremely expensive manufacturing process that is not finically feasible for Solar Turbines. All of these requirements must be met while also increasing the joint stiffness, increasing the lifecycle and reliability of the turbine engine.

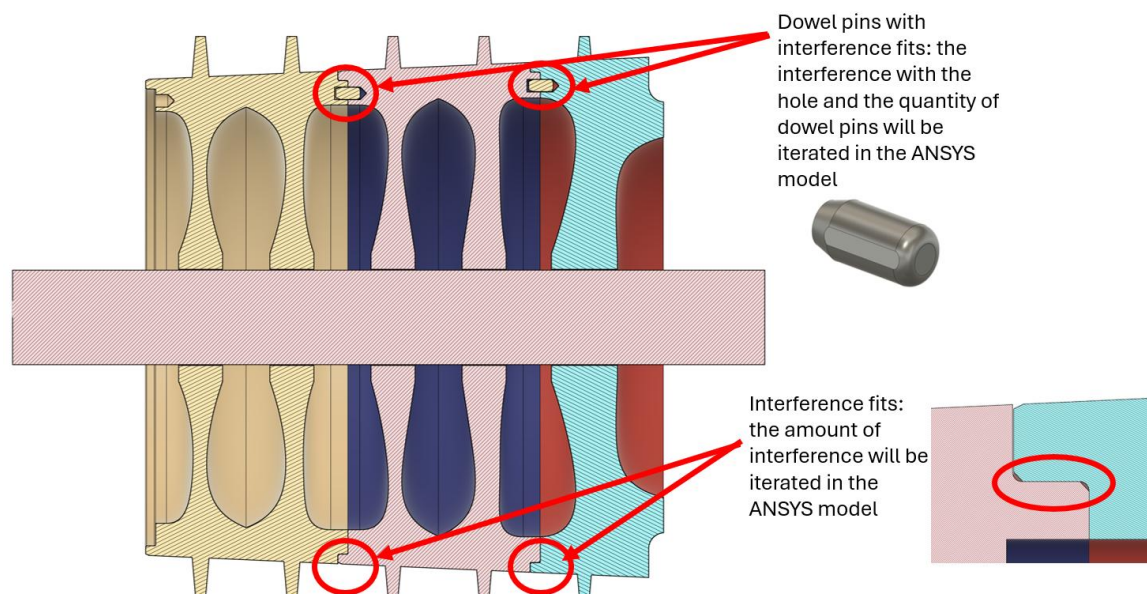


Figure 3. Base configuration mating method used by Solar Turbine's T130.

The first configuration that we will test will comprise of five solid steel dowel pins with an axial and radial interference fit that will be determined later by our Ansys model. This press fit configuration will be accompanied by a pilot fit that closely resembles Solar Turbines' current configuration with altered

interference values, which will also be determined later by our Ansys model. To guarantee that the shear and fatigue requirements associated with our mating joints are met, we will be using Solar Turbines' current dowel pin dimensions. We believe this configuration should be tested because it would provide us with similar data collected by Solar Turbines when they tested their current configuration. This is important because it will assist us by helping identify any issues that might arise with our Ansys software by pointing out any large discrepancies that should not exist. Another reason why this configuration should be tested is to acquire base line results, for which we will use to compare our other configurations. Lastly, this configuration acts as a safe first trial for us to test and ensure our manufacturing and testing processes are accurate and repeatable.

The second configuration that we will test will comprise of seven hardened steel dowel pins with an axial and radial interference fit that will be determined later by our Ansys model. This press fit configuration will also be accompanied by a pilot fit similar to the first configuration but with different interference values. The reason why we believe we should test this configuration is to study how having an increased number of pins and press fit stress along the weldments and hubs affect overall joint stiffness. This will also help us study how having increased dowel pins increased the complexity of manufacturability and assembly due to tolerance propagation. Having several high-tolerance holes becomes difficult to manufacture because slight errors that are within our tolerance callout may affect the precision of our part and affect the joint stiffness in ways we may have not considered. Important factors that need to be studied with this configuration are how having more dowel pins evenly spaced out throughout the pieces affects the stress distribution within the mating parts. This could have unforeseen implications by increasing the overall stress in the part and inducing premature stress cracking that is not experienced with the standard five pin configuration. Lastly, for this configuration we will also need to keep or exceed the same cross-sectional area that the original five dowel pins had but with seven dowel pins. This can be accomplished by downsizing the current dowel pins sizes to be comparable to a 7-pin configuration.

For our last configuration, we will use a fully welded assembly. This involves creating a model of the aft hub and aft and forward weldments that will be joined only by an outer weld where the pilot fit currently exists. The reason why we will be testing this configuration is to measure how a permanent bond between the two mating parts performs in comparison to our dowel pin and pilot joint configurations. The reason why we are using a welded joint configuration is because welded joints offer one of the highest joint stiffnesses possible for joining parts together. This will help us gauge how successful our pin configurations are at increasing joint stiffness by comparing it to the theoretical maximum joint stiffness. The maximum joint stiffness occurs when the two parts are fused together into one piece, so the stiffness is dependent on material and geometrical properties. The issue with making this configuration purely welded is that it is difficult and costly to fully weld the aft hub with the aft weldment, as both are steel. In addition, other issues associated with this process are potential warping due to heat concentrations which are difficult to correct. This is very detrimental to the assembly because it is rotating at high speed where any imbalance or warping in the part compounds exponentially, making the assembly unstable and prone to premature failure, which is unacceptable due to the turbine's long life requirement.

Alternative Solutions Table

If the chosen concept is not approved by the customer, there are several alternative options the team can propose. First, the number of dowel pins can be easily adjusted. Initially, there will be proposed configurations with five and seven dowel pins to hold the rotor plates together. If the number of dowel pins is too high, the CAD models can be modified to match the customer's recommendations and needs. Another concept that can be proposed is the use of different materials for the rotor plates. Currently, Solar Turbines uses a nickel alloy to manufacture their rotor plates. However, since the project aims to build a model and procedure that can accurately predict the stiffness of a turbine configuration, using less stiff materials could help produce larger deflections and more easily validate the data. If a material that is too stiff is chosen, the data may be difficult to read because the changes would be smaller. These different concepts are summarized below in Table 1.

The alternative solutions for our project will depend on information we will gain later during the project. For the dowel pins, there are more than just the 4 and 6 pin configurations that can be chosen. However, this design decision will be driven by the ANSYS model. Once the preloads and temperatures are taken into account for the natural frequency, then we will only decide what number of pin configurations we will want to try. For the material choices, we will first propose steel. Choosing steel means the model we build we resemble Solar Turbines' actual prototype more than aluminum or copper would. Steel would also be easier to weld for the last of our tested configurations.

Table 1: Alternative Solutions Considered

Solution Name	Description	Justification
4 dowel pins	4 dowel pins spaced equally about the center of the turbine shaft	Less dowel pins may be preferred to be explored to reduce manufacturing and assembly time
6 dowel pins	6 dowel pins spaced equally about the center of the turbine shaft	More pins can possibly help keep the system stiffer and result in less shifting
Material Choice	Materials like aluminum or copper are less stiff	Choosing a less stiff material can assist in noticing changes in deflection easier to validate the ANSYS model
Rotor Plate Testing	The testing will be completed using assuming there is no shaft that runs through the connected rotor plates	No shaft running through the rotor plates will be reduce the lumped mass component of the analysis and focus more on the rotor plates

Appendix A: Morphological Matrix

The morphological matrix was our second design matrix we used to narrow down ideas on which mating configurations we were going to analyze with the Ansys model. This matrix was used to not only display the various methods two weldments could be mated together, but also display which methods achieved which required design functions. The column on the far left contains the functions developed in our function decomposition, which explain the ways the project achieves its goals. The remaining columns contain the ideas thought of to complete each function.

Table 2: Morphological Matrix

Morphological Matrix					
Function	Option 1	Option 2	Option 3	Option 4	Option 5
Validate the ANSYS model	use another FEA software to perform the same analysis	hand calculations	physical model with different pin radial positions to perform modal impact testing	physical model with different interference fits to perform modal impact testing	physical model with different material variations
Improve axial Positioning	pilot fits	bolts	barbed dowel pins	welding	adhesives
Improve Rotational Positioning	bolts	Connex dowl pins	slotted dowel pins	hardened dowel pins	coiled spring pin
Improve configuration	2 pins	3 pins	4 pins	5 pins	6 pins
Improve installation	bolts	Flat vent dowel pins	adhesive	2 pins	Connex dowel pins
Ease of manufacturability	Hardened dowel pins	Welding	bolts	adhesives	pilot fit
Adjustable geometry	slotted dowel pins	Bolts	adhesive	welding	pilot fits
Joining force	Welding	solid barbed dowel pin	bolts	adhesives	Connex dowel pins
Adjustable interference fit	dowel pin diameters	Bolt diameters	Welding bead width	pilot diameter	adhesive thickness
Prolonged life cycle	heat treatment	Anodization	sacrificial pieces	lubrication	malleable
Inexpensive material selection	Aluminum	stainless steel	Inconel	Cres steel	Nickel Alloy
Affordable production methods	sheet metal	metal 3d printed	casted	machined	Water Jet

Appendix B: Weighted Decision Matrix

The weighted decision matrix was the last matrix used to finalize the configurations we planned to analyze and test. Included below is the weighted decision matrix along with our morphological matrix color coded to show the combinations of ideas we selected to consider for the Ansys modeling. Combined, the two matrices show the ideas we decided were the best to consider. The column on the far left of the weighted decision matrix shows the same functions shown in the morphological matrix, followed by a column of weights for each function, showcasing the importance of each function. From there, each string of color-coded ideas were assessed on how they handled each function. The bottom row contains the total score for each string of ideas, showing the best ideas to the worst.

Table 3: Weighted Decision Matrix

Weighted Decision Matrix					
Team: CCRP	11/9/2025	Idea 1	Idea 2	Idea 3	Idea 4
Specification	Weight	Red	Blue	Green	Orange
Validate the ANSYS model	0.15	5	4	4	2
Improve axial Positioning	0.1	4	3	2	1
Improve Rotational Positioning	0.05	3	3	2	2
Improve configuration	0.1	3	3	4	2
Improve installation	0.05	3	4	3	4
Ease of manufacturability	0.1	3	4	3	3
Adjustable geometry	0.05	5	4	3	3
Joining force	0.15	4	2	3	2
Adjustable interference fit	0.05	5	3	3	1
Prolonged life cycle	0.1	5	2	3	2
Inexpensive material selection	0.05	3	3	3	3
Affordable production methods	0.05	4	4	1	5
Total	1	4	3.15	3	2.3

Appendix C: Preliminary Analysis

Preliminary analysis on the pin and pilot joint loading was completed using hand calculations. This analysis was done to give an idea of the types of forces normally seen in a Solar Turbines' compressor and rotor. Using specific criteria from Solar Turbines, we conducted quick mathematical stress calculations.

We began our basic analysis by calculating the maximum torque and stress that can be translated through the aft hub, aft weldment, and forward weldment. By calculating the max torque, we needed to solve for the max shear force the current 5 dowel pin configuration can handle in addition to the torque that can be carried by the outer pilot fit. To calculate the torque that the outer pilot joint can carry we needed to solve the pressure at the interface between the aft weldment and the aft hub. This was accomplished by identifying the interference fit and material properties of the interfacing parts. Moreover, having calculated the max pressure and torque that can be transmitted through the assembly, we calculated the max stress that the assembly could experience. This helped us determine a more accurate interface configuration that will have to meet or exceed the torque and stress the current configuration can handle.

Preliminary analysis

1

TOPICS: / criteria

shear, press fit, radial stress, frequency;

done

done $\phi = 0.500 \pm 0.001$

length $L = 0.80 \pm 0.02$

done

incont X 75 D AMS 5049
hardness RC 32-39

Pin - aft weld - hub

5 pins

counter bore

hole

hole $\phi = 0.5180 \pm 0.0005$

length $L = 0.50 \pm 0.015$

} aft weld mit

• aft hub

cup

• outer diam = 20.044 ± 0.001

$L = 0.410 \pm 0.002$

done

hole $\phi = 0.4998 \pm 0.003$

length $L = 0.45 \pm 0.1$

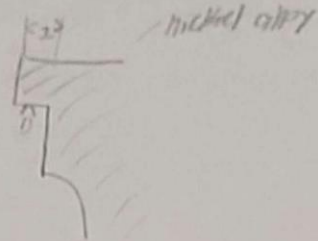
[outer dia = 19.6505 ± 0.0010]

Pin weld

[Diam = 19.684 ± 0.001]

• $F = \sqrt{\frac{h}{m}}$ Tbd

press fits



Doubt Pin Shear

5 pins

$$\text{net area} = 5 \cdot \frac{\pi D^2}{4} ; D_{\max} = 0.5003 \quad D_{\min} = 0.5001$$

$$\text{net area}_{\min} = 0.98214 \text{ in}^2$$

$$\eta = \frac{F}{A} ; \text{Assume Force Perpendicular/along moment}$$



$$\eta_{\max} = 75 \text{ ksi for Inconel}$$

$$F_{\max} = \eta \cdot A_{\min}$$

$$F_{\max} = 73660.5332 \text{ lbf}$$

@ 18.960" radius

$$\text{Torque} = F \cdot r$$

$$\text{Max torque through hub} = 1396603.71 \text{ lbf-in}$$

$$\text{Pins alone} = 116383.642 \text{ lbf-ft}$$

in shear

$$\text{Press fit} \quad P = \frac{8}{d \left[\frac{1 - \gamma_s^2}{E_s} + \frac{1 - \gamma_h^2}{E_h} \cdot \frac{D_o^2 + d^2}{D_o^2 - d^2} \right]}$$

d = shaft diam

D_o = hub's outside diameter in in

γ = interference

Nickel 901 $\rightarrow \gamma = 0.32$

$$E = 200-207 \text{ GPa} = 2.90 \text{ E7} - 3.002 \text{ E7 psi}$$

Same material

• hub to fit smallest γ

hub 20.044 ± 0.001

shaft 20.067 ± 0.001

$$20.045 - 20.066 = [-0.021"]$$

$$P_{\text{hub}} = \frac{0.0d1}{20.066} \left[\frac{1-0.32^2}{2.901E7} + \frac{1-0.32^2}{2.901E7} \cdot \left(\frac{0.0d1^2 + 20.066^2}{d1 - 20.066} \right) \right]$$

$$P_{\text{hub}} = 1470.9 \text{ psi}$$

$$F_{\text{ric}} = N P \cdot A_{\text{cont}}$$

assume A_{cont} = middle ground between 20.045 & 20.066

$$A_{\text{cont}} = L \cdot \pi \cdot d \rightarrow L = 0.410 \pm 0.002 \text{ assume } 0.408$$

$$A = 0.408 \cdot \pi \cdot 20.0555$$

$$A_{\text{cont}} = 25.7065$$

assume $N_{\text{stat}} = 0.15$ dry steel

$$F_{\text{ric}} = 0.15 \cdot 1470.9 \cdot 25.7065 \text{ [psi} \cdot \text{in}^2]$$

$$F_{\text{ric}} = 5671.77 \text{ lbf}$$

$$\text{max torque} = F \cdot \frac{D_{\text{cont}}}{2}$$

$$T_{\text{max}} = 56675.12 \text{ lbf-in}$$

$$= 4739.594 \text{ lbf-ft}$$

Our Gantt chart has been used from the beginning of this project to set out reasonable and actionable goals throughout the design and creation process. The next large checkpoint of this project is the Critical Design Review (CDR). The Gantt chart shown below displays the various tasks and waypoints we will complete from this PDR to the CDR.

